

CS565 Networking Term Project
Sonny Ngo

Access Control Warnings

What are Access Control Warnings?

Unauthorized access control warnings are security alerts prompted when a system detects abnormal or unwarranted attempts to access data and or resources. Said warnings typically stem from intrusion detection systems (IDS), behavioral based analytics, and authentication monitoring. In layman's terms, it is when a person or persons attempt to access a system without permission, thus a user behaves outside their normal pattern. When trying to gain access to these systems, multiple failed login attempts are usually generated.

Example vulnerabilities of Access Control:

1. Faulty Monitoring

Oftentimes with access control security, attacks go unnoticed. Zero day exploits are attacks or vulnerabilities that are unbeknownst to the vendor or developer with no updated or modern day approach/patch to mitigate. This relates to faulty monitoring as threat actors can utilize these exploits to bypass intrusion detection systems that cannot detect these exploits, leading to unnoticed data theft. An example of this could be a brute force, where up to fifty login attempts are tried through brute force but no warning is triggered.

Lab

This lab demonstrates how access control warnings can be generated and analyzed using an intrusion detection system in Kali Linux and where Snort was configured to monitor network traffic.

Step 1: The “*ip a*” command was executed to analyze the system’s network interfaces, allowing for identification of the primary interface (*eth0*) and retrieval of its corresponding IPv4 address for use in traffic monitoring.

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86271sec preferred_lft 86271sec
    inet6 fd00::682f:8313:8786:d4ce/64 scope global dynamic noprefixroute
        valid_lft 86273sec preferred_lft 14273sec
    inet6 fe80::f83f:8ef8:bab7:4df9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

“sudo nano /etc/snort/snort.conf”

Step 2: I used the following command to see if the configuration is valid: “*sudo snort -c /etc/snort/snort.lua -T*”.

Step 3: Then, I opened the local rules file using “*sudo nano /etc/snort/rules/local.rules*” Furthermore, I pasted the following rule demonstrated by the figure w/ the command below.

```
GNU nano 8.7.1 /etc/snort/local.rules
alert icmp any any -> any any ( msg:"Access Control Warning: ICMP Traffic Detected"; sid:100000
```

alert icmp any any -> any any (msg:"Access Control Warning: ICMP Traffic Detected"; sid:1000002; rev:1;)

Step 5: I opened the Snort configuration using the command: “*sudo nano /etc/snort/snort.lua*”, and then used the block below for the IDS.

```
ips =
{
  enable_built_in_rules = true,
  rules = [[
    include /etc/snort/rules/local.rules
  ]]
}
```

Step 6: I revalidated using the command: “*sudo snort -c /etc/snort/snort.lua -T*”.

Step 7: Then, I ran Snort in the first terminal using “*sudo snort -c /etc/snort/snort.lua -i lo -A alert_fast -k none*”.

Step 8: For the ICMP rule, I used “*ping -c 4 127.0.0.1*” in the second terminal to prompt the Access Controlling Warning prompt.

```
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] lo
04/12-17:37:47.184527 [**] [116:151:1] "(decode) same src/dst IP" [**] [Priority: 3] {ICMP} 127.0.0.1 → 127.0.0.1
04/12-17:37:47.184527 [**] [116:150:1] "(decode) loopback IP" [**] [Priority: 3] {ICMP} 127.0.0.1 → 127.0.0.1
04/12-17:37:47.184527 [**] [1:1000002:1] "Access Control Warning: ICMP Traffic Detected" [**] [Priority: 0] {ICMP} 127.0.0.1 → 127.0.0.1
```

First terminal with Snort running, as well as the “Access Control Warnings” prompt

```
➤$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=3.49 ms
```

Second terminal running the ping command

Existing Warning Systems

- Intrusion Detection Systems (IDS)
- Example: Snort
- Monitors network traffic in real time
- Detects suspicious patterns (e.g., repeated access attempts)
- Generates alerts based on predefined rules

Improvement to Warning Systems

- Detects anomalies instead of relying only on rules
- Identifies unusual login patterns or access behavior
- Reduces false positives